

10(a)	spontaneous emission of (ionising) radiation	B1
	emission from unstable nucleus	B1
10(b)(i)	curve with decreasing negative gradient passing through $(0, N_0)$	B1
	curve passing through $(T, 0.5N_0)$	B1
	curve passing through $(2T, 0.25N_0)$ and $(3T, 0.125N_0)$	B1
10(b)(ii)	line through origin with positive gradient	B1
	straight line passing through (N_0, A_0)	B1
10(c)(i)	activity	B1
10(c)(ii)	decay constant	B1
10(d)	$N = N_0 \exp(-\ln 2 \times 1.70T / T)$	C1
	$N / N_0 = 0.31$	A1